

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250287817
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	PENG; Simin

DISPLAY PANEL

Abstract

A display panel is provided, including a substrate, a first inorganic layer, a first organic layer, and an encapsulation layer. The first organic layer includes a planarization layer disposed in a display area and a barrier structure disposed in a peripheral area. The display panel also includes a stacked structure consisting of an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered. There is a gap between the organic sub-layer within the stacked structure and the barrier structure. The encapsulation layer contacts the first inorganic layer at the gap.

Inventors:	PENG; Simin (Wuhan, Hubei, CN)
Applicant:	WUHAN CHINA STAR OPTOELECTRONICS SEMICONDUCTOR DISPLAY TECHNOLOGY CO., LTD. (Wuhan, Hubei, CN)
Family ID:	1000008668946
Assignee:	WUHAN CHINA STAR OPTOELECTRONICS SEMICONDUCTOR DISPLAY TECHNOLOGY CO., LTD. (Wuhan, Hubei, CN)
Appl. No.:	18/861578
Filed (or PCT Filed):	June 29, 2023
PCT No.:	PCT/CN2023/103552

Foreign Application Priority Data

CN	202210956012.6	Aug. 10, 2022
----	----------------	---------------

Publication Classification

Int. Cl.: H10K59/80 (20230101); H10K59/122 (20230101); H10K59/131 (20230101)

Background/Summary

FIELD OF DISCLOSURE

[0001] The present application relates to the field of display technology, and particularly to a display panel.

DESCRIPTION OF RELATED ART

[0002] In the manufacturing process of organic light-emitting display (OLED) panels, to enhance production efficiency and reduce costs, individual OLED panels are typically fabricated on a display mother panel and then cut from it. During the preparation of the display mother panel, the overall thin encapsulation layer is generally an inorganic encapsulation layer prepared by chemical vapor deposition (CVD). Due to the inherent gap between the mask and the display mother panel, reactive gases can infiltrate areas masked by the mask, leading to the formation of shadows in these regions.

[0003] If a shadow falls on the cutting path, it results in an increased thickness of the inorganic film at the cutting location, which can induce edge cracks during the cutting process. These edge cracks have the potential to propagate from the encapsulation layer into the display area, causing panel edge failure. To ensure a high cutting yield, it is common practice to increase the distance between the edge of the CVD deposition area and the cutting path, thereby preventing shadows from overlapping with the cutting path. However, as the bezel width is continuously reduced, shadows inevitably encroach upon the cutting paths. The narrower the bezel, the thicker the shadow becomes, escalating the risk of cutting cracks and significantly impacting the lifespan of the display panel.

SUMMARY OF INVENTION

[0004] One embodiment of the present application provides a display panel designed to address the problem of shadows covering cutting paths in conventional display panels, which leads to the formation of cutting cracks.

[0005] In order to address the above issues, the present application provides the following technical solutions.

[0006] The present application provides a display panel including a display area and at least one peripheral area located on at least one side of the display area, the display panel including: [0007] a substrate; [0008] a first inorganic layer disposed on one side of the substrate; [0009] a first organic layer disposed on one side of the first inorganic layer away from the substrate, wherein the first organic layer includes a planarization layer disposed in the display area and a barrier structure disposed in the peripheral area; and [0010] an encapsulation layer disposed on one side of the first organic layer away from the substrate; [0011] wherein the display panel further includes a stacked structure disposed in the peripheral area and located on one side of the barrier structure away from the display area, the stacked structure including an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered, wherein a gap is defined between the organic sub-layer and the barrier structure, and the encapsulation layer contacts the first inorganic layer at the gap. [0012] In the display panel of the present application, the encapsulation layer includes a first inorganic encapsulation layer, a first organic encapsulation layer, and a second inorganic encapsulation layer, sequentially layered; [0013] wherein the first inorganic encapsulation layer covers the display area and extends to cover the barrier structure, and the first inorganic encapsulation layer contacts the first inorganic layer at the gap.

[0014] In the display panel of the present application, the first organic layer further includes a first organic filler layer disposed on one side of the barrier structure away from the display area; [0015] wherein the stacked structure includes the first inorganic layer, the first organic filler layer, and the first inorganic encapsulation layer, sequentially layered.

[0016] In the display panel of the present application, the first inorganic layer includes at least one groove defined on one side of the barrier structure away from the display area; [0017] wherein the first organic filler layer covers the groove.

[0018] In the display panel of the present application, the first organic layer further includes:

[0019] a pixel definition layer disposed on one side of the planarization layer away from the substrate; and [0020] a plurality of support pillars disposed on one side of the pixel definition layer away from the substrate;

[0021] wherein the first organic filler layer includes a same organic material as one or a combination of the planarization layer, the pixel definition layer, and the support pillars.

[0022] In the display panel of the present application, the stacked structure includes the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; [0023] wherein the second organic filler layer includes a same organic material as the organic encapsulation layer.

[0024] In the display panel of the present application, the display panel further includes: [0025] a second inorganic layer disposed on one side of the encapsulation layer away from the substrate;

and [0026] a third organic filler layer, disposed between the second inorganic layer and the encapsulation layer and located on one side of the barrier structure away from the display area;

[0027] wherein the stacked structure includes the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, the second organic filler layer, the second inorganic encapsulation layer, the third organic filler layer, and the second inorganic layer, sequentially layered, and wherein the third organic filler layer includes a same organic material as the organic encapsulation layer.

[0028] In the display panel of the present application, the stacked structure includes the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; [0029] wherein the second organic filler layer includes a same organic material as the organic encapsulation layer.

[0030] In the display panel of the present application, the first inorganic layer further includes at least one groove defined on one side of the barrier structure away from the display area, and the first organic layer further includes a crack prevention layer disposed between the barrier structure and the stacked structure, the crack prevention layer covering the groove.

[0031] In the display panel of the present application, a thickness of the organic sub-layer is in a range of 2 micrometers to 10 micrometers.

[0032] The present application provides a display panel including a display area and at least one peripheral area located on at least one side of the display area, characterized in that the display panel includes: [0033] a substrate; [0034] a first inorganic layer disposed on one side of the substrate; [0035] a first organic layer disposed on one side of the first inorganic layer away from the substrate, wherein the first organic layer includes a planarization layer disposed in the display area and a barrier structure disposed in the peripheral area; [0036] an encapsulation layer disposed on one side of the first organic layer away from the substrate; and [0037] a driving circuit layer, disposed on one side of the substrate, including a plurality of inorganic film layers and a plurality of organic film layers, wherein the first inorganic layer includes one or more inorganic film layers combined within the driving circuit layer, and the first organic layer includes one or more organic film layers combined within the driving circuit layer; [0038] wherein the display panel further includes a stacked structure disposed in the peripheral area and located on one side of the barrier structure away from the display area, the stacked structure including an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered, wherein a gap is defined

between the organic sub-layer and the barrier structure, and the encapsulation layer contacts the first inorganic layer at the gap.

[0039] In the display panel of the present application, the encapsulation layer includes a first inorganic encapsulation layer, a first organic encapsulation layer, and a second inorganic encapsulation layer, sequentially layered; [0040] wherein the first inorganic encapsulation layer covers the display area and extends to cover the barrier structure, and the first inorganic encapsulation layer contacts the first inorganic layer at the gap.

[0041] In the display panel of the present application, the first organic layer further includes a first organic filler layer disposed on one side of the barrier structure away from the display area; [0042] wherein the stacked structure includes the first inorganic layer, the first organic filler layer, and the first inorganic encapsulation layer, sequentially layered.

[0043] In the display panel of the present application, the first inorganic layer includes at least one groove defined on one side of the barrier structure away from the display area; [0044] wherein the first organic filler layer covers the groove.

[0045] In the display panel of the present application, the first organic layer further includes: [0046] a pixel definition layer disposed on one side of the planarization layer away from the substrate; and [0047] a plurality of support pillars disposed on one side of the pixel definition layer away from the substrate; [0048] wherein the first organic filler layer includes a same organic material as one or a combination of the planarization layer, the pixel definition layer, and the support pillars.

[0049] In the display panel of the present application, the stacked structure includes the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; [0050] wherein the second organic filler layer includes a same organic material as the organic encapsulation layer.

[0051] In the display panel of the present application, the display panel further includes: [0052] a second inorganic layer disposed on one side of the encapsulation layer away from the substrate; and [0053] a third organic filler layer, disposed between the second inorganic layer and the encapsulation layer and located on one side of the barrier structure away from the display area; [0054] wherein the stacked structure includes the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, the second organic filler layer, the second inorganic encapsulation layer, the third organic filler layer, and the second inorganic layer, sequentially layered, and wherein the third organic filler layer includes a same organic material as the organic encapsulation layer.

[0055] In the display panel of the present application, the stacked structure includes the first inorganic encapsulation layer, the second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; [0056] wherein the second organic filler layer includes a same organic material as the organic encapsulation layer.

[0057] In the display panel of the present application, the first inorganic layer further includes at least one groove defined on one side of the barrier structure away from the display area, and the first organic layer further includes a crack prevention layer disposed between the barrier structure and the stacked structure, the crack prevention layer covering the groove.

[0058] In the display panel of the present application, wherein a thickness of the organic sub-layer is in a range of 2 micrometers to 10 micrometers.

BENEFICIAL EFFECTS

[0059] The beneficial effects of this application are as follows: The display panel provided in this embodiment includes a first inorganic layer, a first organic layer, and an encapsulation layer. The first organic layer includes a planarization layer in the display area and a barrier structure in the peripheral area. A stacked structure is formed in the peripheral area, located on one side of the barrier structure away from the display area. The stacked structure includes an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered. A gap is formed between the

organic sub-layer and the barrier structure, allowing the encapsulation layer to contact the first inorganic layer at the gap, thereby forming a complete encapsulation. Within the stacked structure, the organic sub-layer is positioned between two adjacent inorganic sub-layers, creating an inorganic/organic/inorganic alternately-layered structure. This is analogous to dividing a thick inorganic film layer into multiple thinner inorganic sub-layers. This configuration significantly reduces the risk of cracks forming in the stacked structure during the cutting of the display mother panel. This approach enables the realization of ultra-narrow bezels while simultaneously enhancing the performance and longevity of the light-emitting devices in the display panel.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0060] To clarify the technical solutions in the embodiments of the present application more clearly, a brief introduction to the drawings used in the description of the embodiments is provided below. It is apparent that the drawings described below are only some embodiments of this application. Other drawings can also be derived by those skilled in the art without creative effort based on these drawings.

[0061] FIG. 1A is a schematic top view of a display panel provided in an embodiment of the present application.

[0062] FIG. 1B is a schematic top view of a display mother panel provided in an embodiment of the present application.

[0063] FIG. 2A is a first type schematic cross-sectional view taken along line A-A of FIG. 1B.

[0064] FIG. 2B is a first type schematic cross-sectional view of the display panel provided in an embodiment of the present application.

[0065] FIG. 3 is a schematic partial cross-sectional view at point B of FIG. 1B.

[0066] FIG. 4A is a second type schematic cross-sectional view taken along line A-A of FIG. 1B.

[0067] FIG. 4B is a second type schematic cross-sectional view of the display panel provided in an embodiment of the present application.

[0068] FIG. 5A is a third type schematic cross-sectional view taken along line A-A of FIG. 1B.

[0069] FIG. 5B is a third type schematic cross-sectional view of the display panel provided in an embodiment of the present application.

[0070] FIG. 6A is a fourth type schematic cross-sectional view taken along line A-A of FIG. 1B.

[0071] FIG. 6B is a fourth type schematic cross-sectional view of the display panel provided in an embodiment of the present application.

DETAILED DESCRIPTION OF EMBODIMENTS

[0072] In conjunction with the drawings of the embodiments of the present application, the technical solutions in the embodiments of this application are described clearly and completely. Obviously, the embodiments described herein are only a part of the embodiments of this application and not all of them. Based on the embodiments of this application, all other embodiments obtained by those skilled in the art without creative effort fall within the scope of protection of this application. Furthermore, it should be understood that the specific embodiments described here are used only to illustrate and explain this application and are not intended to limit this application. In this application, unless otherwise specified, directional terms such as “up” and “down” usually refer to the orientation of the device in actual use or operation, specifically as indicated in the diagram directions of the drawings; and “inside” and “outside” refer to the contour of the device.

[0073] Please refer to FIGS. 1A and 1B, where FIG. 1A is a schematic top view of a display panel provided in one embodiment of the present application, and FIG. 1B is a schematic top view of a display mother panel provided in an embodiment of this application. This embodiment provides a

display panel **200**, which includes a display area **1a** and a peripheral area **1b** located on at least one side of the display area **1a**.

[0074] The display panel **200** is cut from the display mother panel **100**, as shown in FIG. **1B**. The display mother panel **100** includes a substrate base **300** and a plurality of display panels **200** set on one side of the substrate base **300**, with the display panels **200** being formed by cutting the display mother panel **100**. Specifically, the display mother panel **100** includes a cutting zone **1c** equipped with cutting paths. The display mother panel **100** is cut along these cutting paths to form multiple independent display panels **200**. In this embodiment, the cutting techniques used may include wheel cutting, laser cutting, or a combination of both.

[0075] Please refer to FIGS. **2A** and **2B**. FIG. **2A** is a first type schematic cross-sectional view taken along line A-A of FIG. **1B**, and FIG. **2B** is a first type schematic cross-sectional view of the display panel provided in one embodiment of the present application. FIG. **2A** shows a cross-section of the display mother panel **100** before cutting, while FIG. **2B** shows a cross-section of the display panel **200** after cutting.

[0076] The display panel **200** includes a substrate **204**, a first inorganic layer **201**, a first organic layer **202**, and an encapsulation layer **23**. The first inorganic layer **201** is positioned on one side of the substrate **204**. The first organic layer **202** is set on one side of the first inorganic layer **201** that is away from the substrate **204** and includes a planarization layer set in the display area **1a** and a barrier structure **114** set in the peripheral area **1b**. The encapsulation layer **23** is positioned on one side of the first organic layer **202** that is away from the substrate **204**.

[0077] The display panel **200** also includes a stacked structure **10** located in the peripheral area **1b** and on one side of the barrier structure **114** that is away from the display area **1a**. The stacked structure **10** consists of an inorganic sub-layer **10a**, an organic sub-layer **11a**, and an inorganic sub-layer **10a** sequentially layered. There is a gap between the organic sub-layer **11a** and the barrier structure **114**, where the encapsulation layer **23** contacts the first inorganic layer **201** at the gap.

[0078] It is understandable that as bezels are progressively reduced, shadows increasingly encroach upon the cutting paths. The narrower the bezel, the thicker the shadow, resulting in a thicker inorganic stack in the cutting zone **1c**. This, in turn, elevates the risk of cutting cracks. The present application addresses this challenge by introducing an organic sub-layer **11a** into the stacked structure **10**. This organic sub-layer **11a** is positioned between adjacent inorganic sub-layers **10a**, creating a multi-layered inorganic/organic/inorganic structure. Effectively, this divides the combined inorganic film layers into multiple thinner inorganic sub-layers **10a**, significantly mitigating the risk of cracks during the cutting process of the display mother panel **100**. This approach not only enables the realization of ultra-narrow bezels but also enhances the performance and longevity of the light-emitting devices within the display panel **200**.

[0079] It should be noted that in FIG. **2B**, the display area **1a** and the peripheral area **1b** are illustratively differentiated by dashed lines. The display area **1a** accommodates light-emitting units and pixel circuits that drive these light-emitting units, while the peripheral area **1b** serves as a bezel region of the display panel **200**, housing the driving circuits and various signal lines.

[0080] The substrate **204** and the substrate base **300** can be the same film layer, with the substrate **204** being cut from the substrate base **300**. In this embodiment, the substrate **204** can be a flexible substrate. The flexible substrate may be composed of a single or multiple layers of flexible organic films, which can resist stress impacts to some extent, helping to reduce the risk of cracks in the stacked structure **10** on the substrate **204** during the cutting process.

[0081] Optionally, the flexible substrate can be made from materials such as polyimide (PI), polyethylene terephthalate (PET), polybutylene naphthalate (PBN), or polycarbonate. In this embodiment, the flexible substrate uses a single layer of PI film. In other embodiments of the present application, the flexible substrate can be a multi-layer PI film, a multi-layer PET film, or have a multi-layer structure with alternating layers of PI and PET films.

[0082] The encapsulation layer **23** serves to protect the light-emitting units and is implemented

using a thin-film encapsulation method. This encapsulation layer **23** can employ a multi-layered inorganic/organic/inorganic structure. For example, in this embodiment, the encapsulation layer **23** includes a first inorganic encapsulation layer **231**, a second inorganic encapsulation layer **232**, and an organic encapsulation layer **233**, sequentially stacked away from the substrate **204**. Both the first and second inorganic encapsulation layers **231** and **232** extend from the display area **1a** to cover the peripheral area **1b**, while the organic encapsulation layer **233** covers the display area **1a**.

[0083] The first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232** extend from the display area **1a** to the peripheral area **1b** and cover the barrier structure **114**. The first inorganic encapsulation layer **231** contacts the first inorganic layer **201** at the gap, thus forming a complete encapsulation. This arrangement provides encapsulation protection for the wiring and other components in the peripheral area **1b**, primarily acting to block the ingress of water and oxygen, thereby enhancing the effectiveness of the encapsulation.

[0084] Optionally, the first and second inorganic encapsulation layers **231** and **232** can be fabricated from materials such as nitrides, oxides, oxynitrides, nitrates, carbides, or any combination thereof.

[0085] The first organic encapsulation layer **233** typically terminates at the barrier structure **114**, serving a role in auxiliary encapsulation and planarization. Optionally, the first organic encapsulation layer **233** can be made from materials such as acrylonitrile, hexamethyldisiloxane, polyacrylates, polycarbonates, polystyrene, or similar materials.

[0086] In this embodiment of the application, the first organic layer **202** further includes a first organic filler layer **111** positioned on one side of the barrier structure **114** away from the display area **1a**. In this configuration, the stacked structure **10** includes the first inorganic layer **201**, the first organic filler layer **111**, and the first inorganic encapsulation layer **231**, sequentially layered.

[0087] It is understood that in conventional technologies, the first inorganic layer **201** and the first inorganic encapsulation layer **231** in the peripheral area **1b** are in direct contact, with the two inorganic layers stacked together. This is equivalent to having a single, thicker inorganic film layer, which increases the risk of cracking during the cutting process. This application addresses this by introducing the first organic filler layer **111** between the first inorganic layer **201** and the first inorganic encapsulation layer **231**. This separates the first inorganic layer **201** and the first inorganic encapsulation layer **231**, preventing direct contact. This is analogous to splitting the inorganic stack formed by the first inorganic layer **201** and the first inorganic encapsulation layer **231**, resulting in an inorganic/organic/inorganic film structure. Essentially, this divides a thicker inorganic film layer into two thinner inorganic sub-layers **10a**, thereby reducing the risk of cracks during cutting.

[0088] It should be noted that the first organic filler layer **111** is an organic film layer that serves as a stress buffer, further reducing the risk of cracking during the cutting process.

[0089] Moreover, the first inorganic layer **201** includes at least one groove **12**, located on one side of the barrier structure **114** away from the display area **1a**, which is covered by the first organic filler layer **111**. This groove **12** acts to block cutting cracks from extending from the peripheral area **1b** to the display area **1a**, thus lowering the risk of failure in the light-emitting units located in the display area **1a**.

[0090] It should also be mentioned that the thickness of each organic sub-layer **11a** ranges from 2 micrometers to 10 micrometers. The reason for setting this thickness range is that on one hand, the organic sub-layer **11a** needs to function as a buffer, thus its thickness should not be too thin; on the other hand, if the organic sub-layer **11a** is too thick during the cutting of the display mother panel **100**, it tends to be prone to adhesion. Preferably, the thickness of each organic sub-layer **11a** ranges from 5 micrometers to 10 micrometers.

[0091] Specifically, please refer to FIG. 3, which is a schematic partial cross-sectional view at point B of FIG. 1B. The display panel **200** includes a driving circuit layer **20** and a light-emitting device layer **21**. The driving circuit layer **20** is located on one side of the substrate **204**, while the light-

emitting device layer **21** is positioned on one side of the driving circuit layer **20** that is away from the substrate **204** and is located in the display area **1a**. The encapsulation layer **23** covers one side of the light-emitting device layer **21** that is away from the substrate **204**. The light-emitting device layer **21** includes multiple light-emitting units, and the driving circuit layer **20** includes multiple pixel driving circuits that are used to drive the light-emitting units. The pixel driving circuits include multiple thin-film transistors.

[0092] The driving circuit layer **20** includes inorganic film layers and organic film layers.

Specifically, the driving circuit layer **20** can include multiple inorganic film layers and multiple organic film layers. In this embodiment of the application, the first inorganic layer **201** can be one of the inorganic film layers in the driving circuit layer **20**, or it can be a combination of multiple inorganic film layers. Similarly, the first organic layer **202** can be one of the organic film layers in the driving circuit layer **20**, or a combination of multiple organic film layers.

[0093] In this embodiment, the thin-film transistors can be of various structures, such as etch stop type, back channel etched type, or top gate type, with no specific limitations.

[0094] As shown in FIG. **3**, in this embodiment of the application, the thin-film transistors are exemplified as top-gate thin-film transistors. The driving circuit layer **20** includes a semiconductor layer **101**, a first gate insulating layer **102**, a first gate layer **103**, a second gate insulating layer **104**, a second gate layer **105**, an interlayer dielectric **106**, a first source-drain metal layer **107**, a first planarization layer **108**, a second source-drain metal layer **109**, and a second planarization layer **110**.

[0095] The semiconductor layer **101** is located on one side of the substrate **204**, the first gate insulating layer **102** covers one side of the semiconductor layer **101** that is away from the substrate **204**, the first gate layer **103** is positioned on one side of the first gate insulating layer **102** that is away from the substrate **204**, the second gate insulating layer **104** covers one side of the first gate layer **103** that is away from the substrate **204**, the second gate layer **105** is disposed on one side of the second gate insulating layer **104** that is away from the substrate **204**, the interlayer dielectric **106** covers one side of the second gate layer **105** that is away from the substrate **204**, the first source-drain metal layer **107** is located on one side of the interlayer dielectric **106** that is away from the substrate **204**, the first planarization layer **108** covers one side of the first source-drain metal layer **107** away from the substrate **204**, the second source-drain metal layer **109** is positioned on one side of the first source-drain metal layer **107** that is away from the substrate **204**, and the second planarization layer **110** covers one side of the second source-drain metal layer **109** away from the substrate **204**.

[0096] In this embodiment, the first inorganic layer **201** can include one or more layers selected from the first gate insulating layer **102**, the second gate insulating layer **104**, and the interlayer dielectric **106**. Similarly, the first organic layer **202** can include one or more layers combined from the first planarization layer **108** and the second planarization layer **110**.

[0097] Specifically, the light-emitting device layer **21** includes a pixel definition layer **211**. The pixel definition layer **211** is positioned on one side of the second planarization layer **110** that is away from the substrate **204**, and the pixel definition layer **211** is an organic film. The light-emitting device layer **21** also includes an anode layer **212**, a light-emitting functional layer **213**, and a cathode layer (not shown in the drawing). The pixel definition layer **211** defines a plurality of pixel openings, and the light-emitting functional layer **213** is located within the pixel openings. The light-emitting functional layer **213** can also include an electron injection layer, an electron transport layer, a light-emitting layer, a hole transport layer, and a hole injection layer, all situated between the anode layer **212** and the cathode layer.

[0098] The display panel **200** also includes a plurality of support pillars **22**. The support pillars are located on one side of the pixel definition layer **211** that is away from the substrate **204** and disposed in the display area **1a**. The material of the support pillars **22** is an organic material.

[0099] In this embodiment, the first organic filler layer **111** can include the same material as any

one or more layers among the planarization layer, the pixel definition layer **211**, and the support pillars **22**. That is, the first organic filler layer **111** can be manufactured using the same process as any one or more of the planarization layer, the pixel definition layer **211**, and the support pillars **22**, which is beneficial for reducing the number of manufacturing steps.

[0100] Specifically, the first organic filler layer **111** includes the same material used in any one or more of the following layers: the first planarization layer **108**, the second planarization layer **110**, the pixel definition layer **211**, and the support pillars **22**.

[0101] In one embodiment, please refer to FIGS. **4A** and **4B**. FIG. **4A** is a second type schematic cross-sectional view taken along line A-A of FIG. **1B**, and FIG. **4B** is a second type schematic cross-sectional view of the display panel provided in one embodiment of the present application. The difference between FIGS. **4A** and **4B** compared to FIGS. **2A** and **2B** is that the stacked structure **10** also includes a second organic filler layer **112** positioned between the first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232**.

[0102] In this case, the stacked structure **10** consists of sequentially layered components: the first inorganic layer **201**, the first organic filler layer **111**, the first inorganic encapsulation layer **231**, the second organic filler layer **112**, and the second inorganic encapsulation layer **232**.

[0103] It is understandable that in the peripheral area **1b**, if the first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232** were in direct contact, these two film layers would be equivalent to a single, thicker inorganic film layer. This would increase the risk of cracks forming during the cutting process. By introducing the second organic filler layer **112**, this application ensures that the first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232** are spaced apart, preventing direct contact. This is analogous to splitting the thick inorganic film layer formed by the stacking of the first and second inorganic encapsulation layers **231** and **232**. Consequently, the stacked structure **10** forms an inorganic/organic/inorganic/organic/inorganic layered structure, which reduces the risk of cracks during cutting.

[0104] Similarly, the second organic filler layer **112** is an organic film layer that acts as a stress buffer, further reducing the risk of cracks during the cutting process.

[0105] In this embodiment, the second organic filler layer **112** can consist of the same organic material as the organic encapsulation layer **233**. Structurally, the second organic filler layer **112** can be set in the same layer as the organic encapsulation layer **233**; from a manufacturing process perspective, the second organic filler layer **112** and the organic encapsulation layer **233** are formed using the same process, which is advantageous for reducing steps and lowering costs.

[0106] In one embodiment, please refer to FIGS. **5A** and **5B**. FIG. **5A** is a third type schematic cross-sectional view taken along line A-A of FIG. **1B**, and FIG. **5B** is a third type schematic cross-sectional view of the display panel provided in one embodiment of the present application. The difference between FIGS. **5A** and **5B** compared to FIGS. **4A** and **4B** is that the display panel **200** additionally includes a second inorganic layer **203** and a third organic filler layer **113**. The second inorganic layer **203** is positioned on one side of the encapsulation layer **23** that is away from the substrate **204**, and the third organic filler layer **113** is disposed between the second inorganic layer **203** and the encapsulation layer **23**, and is located on one side of the barrier structure **114** away from the display area **1a**.

[0107] In this scenario, the stacked structure **10** consists of sequentially layered components: the first inorganic layer **201**, the first organic filler layer **111**, the first inorganic encapsulation layer **231**, the second organic filler layer **112**, the second inorganic encapsulation layer **232**, the third organic filler layer **113**, and the second inorganic layer **203**.

[0108] It is understandable that in the peripheral area **1b**, if the second inorganic encapsulation layer **232** and the second inorganic layer **203** were in direct contact, these two layers would be equivalent to a single, thicker inorganic film layer. This would increase the risk of cracks forming during the cutting process. This application mitigates this risk by incorporating the third organic

filler layer **113**, which ensures that the second inorganic encapsulation layer **232** and the second inorganic layer **203** are spaced apart, preventing direct contact. This is analogous to splitting the thick inorganic film layer formed by the stacking of the second inorganic encapsulation layer **232** and the second inorganic layer **203**. Consequently, the stacked structure **10** forms an inorganic/organic/inorganic/organic/inorganic/organic/inorganic layered structure, reducing the risk of cracks during cutting.

[0109] Similarly, the third organic filler layer **113** is an organic film layer that acts as a stress buffer, further reducing the risk of cracks forming during the cutting process.

[0110] Please refer again to FIG. **3**. The display panel **200** also includes a touch layer **24**. The touch layer **24** is positioned on one side of the encapsulation layer **23** that is away from the substrate **204**. The touch layer **24** includes a first insulation layer **241**, touch electrodes **242**, and a second insulation layer **243**. The first insulation layer **241** extends from the display area **1a** and covers the peripheral area **1b**. The touch electrodes **242** are located on one side of the first insulation layer **241** that is away from the substrate **204** and situated in the display area **1a**. The second insulation layer **243** covers one side of the touch electrodes **242** that is away from the substrate **204**.

[0111] The material for the first insulation layer **241** is an inorganic material, and in this embodiment, the second inorganic layer **203** can include the first insulation layer **241**.

[0112] In the present embodiment, the third organic filler layer **113** can include the same organic material as the organic encapsulation layer **233**.

[0113] In one embodiment, please refer to FIGS. **6A** and **6B**. FIG. **6A** is a fourth type schematic cross-sectional view taken along line A-A of FIG. **1B**, and FIG. **6B** is a fourth type schematic cross-sectional view of the display panel provided in one embodiment of the present application. The difference between FIGS. **6A** and **6B** and FIGS. **2A** and **2B** is that the second organic filler layer **112** is positioned between the first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232**. In this case, the stacked structure **10** includes sequentially layered components: the first inorganic encapsulation layer **231**, the second organic filler layer **112**, and the second inorganic encapsulation layer **232**.

[0114] It is understandable that the second organic filler layer **112** is positioned between the first inorganic encapsulation layer **201** and the first inorganic encapsulation layer **231**. This is analogous to splitting the thicker inorganic film layer formed by the stacking of the first inorganic encapsulation layer **231** and the second inorganic encapsulation layer **232**, resulting in an inorganic/organic/inorganic layered structure within the stacked structure **10**. This configuration mitigates the risk of cracks during the cutting process.

[0115] In this embodiment, the second organic filler layer **112** can be composed of the same organic material as the organic encapsulation layer **233**.

[0116] Furthermore, in this embodiment, the first inorganic layer **201** also includes at least one groove **12** on one side of the barrier structure **114** away from the display area **1a**. The first organic layer **202** includes a crack prevention layer **14** positioned between the barrier structure **114** and the stacked structure **10**. The crack prevention layer **14** covers the groove **12**.

[0117] In this embodiment, the first inorganic encapsulation layer **231** covers the crack prevention layer **14**, and the crack prevention layer **14** and the second organic filler layer **112** are separated by the first inorganic encapsulation layer **231**.

[0118] Beneficial effects: The display panel provided in this embodiment includes a first inorganic layer, a first organic layer, and an encapsulation layer. The first organic layer includes a planarization layer in the display area and a barrier structure in the peripheral area. A stacked structure is formed in the peripheral area, located on one side of the barrier structure away from the display area. The stacked structure includes an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered. A gap is formed between the organic sub-layer and the barrier structure, allowing the encapsulation layer to contact the first inorganic layer at the gap, thereby forming a complete encapsulation. Within the stacked structure, the organic sub-layer is

positioned between two adjacent inorganic sub-layers, creating an inorganic/organic/inorganic alternately-layered structure. This is analogous to dividing a thick inorganic film layer into multiple thinner inorganic sub-layers. This configuration significantly reduces the risk of cracks forming in the stacked structure during the cutting of the display mother panel. This approach enables the realization of ultra-narrow bezels while simultaneously enhancing the performance and longevity of the light-emitting devices in the display panel.

[0119] In summary, although the present application has been disclosed in the preferred embodiments as described above, the preferred embodiments are not intended to limit the application. Those skilled in the art can make various modifications and refinements without departing from the spirit and scope of this application. Therefore, the scope of protection of this application is defined by the scope of the claims.

Claims

1. A display panel comprising a display area and at least one peripheral area located on at least one side of the display area, the display panel comprising: a substrate; a first inorganic layer disposed on one side of the substrate; a first organic layer disposed on one side of the first inorganic layer away from the substrate, wherein the first organic layer comprises a planarization layer disposed in the display area and a barrier structure disposed in the peripheral area; and an encapsulation layer disposed on one side of the first organic layer away from the substrate; wherein the display panel further comprises a stacked structure disposed in the peripheral area and located on one side of the barrier structure away from the display area, the stacked structure comprising an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered, wherein a gap is defined between the organic sub-layer and the barrier structure, and the encapsulation layer contacts the first inorganic layer at the gap.
2. The display panel according to claim 1, wherein the encapsulation layer comprises a first inorganic encapsulation layer, a first organic encapsulation layer, and a second inorganic encapsulation layer, sequentially layered; wherein the first inorganic encapsulation layer covers the display area and extends to cover the barrier structure, and the first inorganic encapsulation layer contacts the first inorganic layer at the gap.
3. The display panel according to claim 2, wherein the first organic layer further comprises a first organic filler layer disposed on one side of the barrier structure away from the display area; wherein the stacked structure comprises the first inorganic layer, the first organic filler layer, and the first inorganic encapsulation layer, sequentially layered.
4. The display panel according to claim 3, wherein the first inorganic layer comprises at least one groove defined on one side of the barrier structure away from the display area; wherein the first organic filler layer covers the groove.
5. The display panel according to claim 3, wherein the first organic layer further comprises: a pixel definition layer disposed on one side of the planarization layer away from the substrate; and a plurality of support pillars disposed on one side of the pixel definition layer away from the substrate; wherein the first organic filler layer comprises a same organic material as one or a combination of the planarization layer, the pixel definition layer, and the support pillars.
6. The display panel according to claim 3, wherein the stacked structure comprises the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; wherein the second organic filler layer comprises a same organic material as the organic encapsulation layer.
7. The display panel according to claim 6, further comprising: a second inorganic layer disposed on one side of the encapsulation layer away from the substrate; and a third organic filler layer, disposed between the second inorganic layer and the encapsulation layer and located on one side of the barrier structure away from the display area; wherein the stacked structure comprises the first

inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, the second organic filler layer, the second inorganic encapsulation layer, the third organic filler layer, and the second inorganic layer, sequentially layered, and wherein the third organic filler layer comprises a same organic material as the organic encapsulation layer.

8. The display panel according to claim 2, wherein the stacked structure comprises the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; wherein the second organic filler layer comprises a same organic material as the organic encapsulation layer.

9. The display panel according to claim 8, wherein the first inorganic layer further comprises at least one groove defined on one side of the barrier structure away from the display area, and the first organic layer further comprises a crack prevention layer disposed between the barrier structure and the stacked structure, the crack prevention layer covering the groove.

10. The display panel according to claim 1, wherein a thickness of the organic sub-layer is in a range of 2 micrometers to 10 micrometers.

11. A display panel comprising a display area and at least one peripheral area located on at least one side of the display area, the display panel comprising: a substrate; a first inorganic layer disposed on one side of the substrate; a first organic layer disposed on one side of the first inorganic layer away from the substrate, wherein the first organic layer comprises a planarization layer disposed in the display area and a barrier structure disposed in the peripheral area; an encapsulation layer disposed on one side of the first organic layer away from the substrate; and a driving circuit layer, disposed on one side of the substrate, comprising a plurality of inorganic film layers and a plurality of organic film layers, wherein the first inorganic layer comprises one or more inorganic film layers combined within the driving circuit layer, and the first organic layer comprises one or more organic film layers combined within the driving circuit layer; wherein the display panel further comprises a stacked structure disposed in the peripheral area and located on one side of the barrier structure away from the display area, the stacked structure comprising an inorganic sub-layer, an organic sub-layer, and an inorganic sub-layer, sequentially layered, wherein a gap is defined between the organic sub-layer and the barrier structure, and the encapsulation layer contacts the first inorganic layer at the gap.

12. The display panel according to claim 11, wherein the encapsulation layer comprises a first inorganic encapsulation layer, a first organic encapsulation layer, and a second inorganic encapsulation layer, sequentially layered; wherein the first inorganic encapsulation layer covers the display area and extends to cover the barrier structure, and the first inorganic encapsulation layer contacts the first inorganic layer at the gap.

13. The display panel according to claim 12, wherein the first organic layer further comprises a first organic filler layer disposed on one side of the barrier structure away from the display area; wherein the stacked structure comprises the first inorganic layer, the first organic filler layer, and the first inorganic encapsulation layer, sequentially layered.

14. The display panel according to claim 13, wherein the first inorganic layer comprises at least one groove defined on one side of the barrier structure away from the display area; wherein the first organic filler layer covers the groove.

15. The display panel according to claim 13, wherein the first organic layer further comprises: a pixel definition layer disposed on one side of the planarization layer away from the substrate; and a plurality of support pillars disposed on one side of the pixel definition layer away from the substrate; wherein the first organic filler layer comprises a same organic material as one or a combination of the planarization layer, the pixel definition layer, and the support pillars.

16. The display panel according to claim 13, wherein the stacked structure comprises the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, a second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; wherein the second organic filler layer comprises a same organic material as the organic encapsulation layer.

17. The display panel according to claim 16, further comprising: a second inorganic layer disposed on one side of the encapsulation layer away from the substrate; and a third organic filler layer, disposed between the second inorganic layer and the encapsulation layer and located on one side of the barrier structure away from the display area; wherein the stacked structure comprises the first inorganic layer, the first organic filler layer, the first inorganic encapsulation layer, the second organic filler layer, the second inorganic encapsulation layer, the third organic filler layer, and the second inorganic layer, sequentially layered, and wherein the third organic filler layer comprises a same organic material as the organic encapsulation layer.

18. The display panel according to claim 12, wherein the stacked structure comprises the first inorganic encapsulation layer, the second organic filler layer, and the second inorganic encapsulation layer, sequentially layered; wherein the second organic filler layer comprises a same organic material as the organic encapsulation layer.

19. The display panel according to claim 18, wherein the first inorganic layer further comprises at least one groove defined on one side of the barrier structure away from the display area, and the first organic layer further comprises a crack prevention layer disposed between the barrier structure and the stacked structure, the crack prevention layer covering the groove.

20. The display panel according to claim 11, wherein a thickness of the organic sub-layer is in a range of 2 micrometers to 10 micrometers.
